Ruitang Yang, PhD

Glaciologist | Glacier Modelling, Data Assimilation and Satellite Remote Sensing

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ACADEMIC PROFILE

Glaciologist and researcher at the University of Oslo specializing in glacier modelling, data assimilation, and satellite remote sensing. Integrates process-based models, Earth-observation data, and Bayesian inference to quantify glacier change and produce uncertainty-aware projections from regional to global scales.

CURRENT APPOINTMENT

2022–Present	**Researcher** | University of Oslo | Norway Leads the development of a coupled glacier-modelling and data-assimilation framework integrating process-based models, Earth-observation data, Bayesian calibration, and ensemble uncertainty analysis.

EDUCATION

2016–2022	**PhD in Physical Geography** | University of Chinese Academy of Sciences | China Thesis: Glacier Change on the Kenai Peninsula, Alaska, Using Remote Sensing and Modelling. Visiting researcher at the University of Alaska Fairbanks, USA (18 months), and the University of Oslo, Norway (12 months)
2012–2015	**MSc in Mathematics Education** | Shaanxi Normal University | China
2007–2011	**BSc in Mathematics and Applied Mathematics** | Shaanxi Normal University | China

SELECTED RESEARCH CONTRIBUTIONS

- **Marine-terminating glacier modelling.** Led development of a coupled OGGM–PyGEM–SERMeQ framework integrating climatic mass balance, glacier geometry, ice dynamics, and frontal ablation. Demonstrated for 71 marine-terminating glaciers in Svalbard using joint Bayesian calibration, multiple observational constraints, and uncertainty propagation through 2100.
- **Kenai Peninsula glacier change.** Developed multi-decadal satellite Earth-observation and modelling workflows to quantify changes in glacier area, mass, surface velocity, and glacier–runoff relationships across the Kenai Peninsula, Alaska, using observations from 1986–2019. The work resulted in three first-author publications and three public regional datasets.

SELECTED PUBLICATIONS

- **Yang, R.**, Ultee, L., Aalstad, K., Debolskiy, M. V., Hock, R., Schmitt, P., Rounce, D., and Li, T. (2026). Joint Bayesian Calibration of Frontal Ablation and Surface Mass Balance in Global Glacier Models. *EGUsphere [preprint]*. doi:10.5194/egusphere-2026-1081
- **Yang, R.**, Hock, R., Rounce, D. R., and Kang, S. (2025). Regional-scale response of glacier speed to seasonal runoff variations on the Kenai Peninsula, Alaska. *Geophysical Research Letters*. doi:10.1029/2025GL115248
- **Yang, R.**, Hock, R., Kang, S., Guo, W., Shangguan, D., Jiang, Z., and Zhang, Q. (2022). Glacier surface speed variations on the Kenai Peninsula, Alaska, 2014–2019. *Journal of Geophysical Research: Earth Surface*. doi:10.1029/2022JF006599
- **Yang, R.**, Hock, R., Kang, S., Shangguan, D., and Guo, W. (2020). Glacier mass and area changes on the Kenai Peninsula, Alaska, 1986–2016. *Journal of Glaciology*. doi:10.1017/jog.2020.32
- Aalstad, K., Alonso-González, E., Pirk, N., Westermann, S., Willmes, C., and **Yang, R.** (2026). Evolving beyond collapse: An adaptive particle batch smoother for cryospheric data assimilation. *arXiv preprint*. arXiv:2601.20049

Complete publication list: Google Scholar

SELECTED RESEARCH SOFTWARE AND DATASETS

- **Coupled Frontal-Ablation Modelling Framework.** Lead designer and developer of an observation-constrained research software framework for marine-terminating glaciers. doi:10.5281/zenodo.18761729
- **Three lead-author Kenai Peninsula datasets.** Glacier mass and area change (1986–2016), velocity maps (2014–2019), and regional surface-speed products: PANGAEA 965738 · PANGAEA 971786 · Mendeley Data

TECHNICAL EXPERTISE

- **Programming and computing:** Python, Shell, Git, Linux, HPC; MATLAB and C
- **Scientific modelling and inference:** OGGM, PyGEM, SERMeQ; process-based modelling, Bayesian inference, data assimilation, ensemble simulation, model calibration, and uncertainty quantification
- **Geospatial and Earth observation:** QGIS, ArcGIS, ENVI, GAMMA; optical, SAR, and laser-altimetry data
- **Research practice:** reproducible workflows, multi-source data integration, model evaluation, open-source software, and research-data publication

AWARDS AND RECOGNITION

2024 — **Top Cited Article 2022–2023**
Journal of Geophysical Research: Earth Surface

2021 — **Best Student Oral Presentation**
International Glaciological Society Nordic Branch Symposium | Oslo

SELECTED PROFESSIONAL SERVICE

2022–Present — **Member and subgroup co-chair, GlacierMIP**
Co-leading development of standardized experimental protocols for GlacierMIP4

2023–Present — **Peer reviewer**
Journal of Glaciology; Advances in Climate Change Research; GIScience & Remote Sensing; AGU Outstanding Student Presentation Award program

2025 — **Organizing committee member**
CryoDA Workshop and Glacier Modelling Workshop | University of Oslo

2022–2023 — **Member, Working Group on Regional Assessments of Glacier Mass Change (RAGMAC)**
International Association of Cryospheric Sciences

SELECTED PRESENTATIONS

2026 — **Modeling Frontal Ablation in Global Glacier Models**
EGU General Assembly | Vienna · PICO

2024 — **Regional-Scale Response of Glacier Speed to Seasonal Runoff Variations on the Kenai Peninsula, Alaska**
AGU Fall Meeting | Washington, D.C. · Poster

2021 — **Glacier Changes on the Kenai Peninsula, Alaska: Mass Balance and Ice Flow Using Remote Sensing**
International Glaciological Society Nordic Branch Symposium | Oslo · Oral

TEACHING AND MENTORING

2011–2016 — **Senior High School Mathematics Teacher and Class Adviser**
Longxi No. 1 Middle School | China
Five years of mathematics teaching; served for three years as class adviser with responsibility for approximately 60 students.

2017–2018 — **Co-mentor, MSc thesis in glacier remote sensing**
University of Chinese Academy of Sciences | China

SELECTED FIELD EXPERIENCE

2018 & 2020 — **Field Assistant, Kennicott Glacier, Alaska, USA**
Ablation-stake drilling, AWS and time-lapse camera installation, and snow and water sampling; four weeks across two campaigns.

2019 — **Field Assistant, Halong Glacier, Qinghai–Tibet Plateau, China**
Ablation-stake drilling, AWS installation; three weeks

2017 — **Snow and UAV Field Campaign, Zhangjiakou, China**
Snow-pit measurements, melt-stake drilling, time-lapse camera installation, and UAV surveys; 66 days